

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Student Name	Divyansh Badaya
SRN	PES1UG24AM094
Project	Digital Campus Library Reservation Gateway
Methodology	Agile Scrum
Sprint Duration	1 Week per Sprint

Purpose

Convert the Lab 1 requirements for the Digital Campus Library Reservation Gateway into Agile Epics and User Stories, prioritize and estimate the backlog, and simulate two one-week sprints in Jira.

1. Epics

Epic	Description
Book Reservation & Holding Queue	Manage book hold requests and maintain a fair FIFO holding queue for members.
Reservation Notifications	Notify members when reserved books become available and when queue changes affect their p
Fine Management	Automatically calculate and maintain overdue fines for borrowed library books.
Hold Cancellation	Allow members to cancel pending reservations while maintaining the correct FIFO queue order.
Librarian Queue Administration	Allow the Head Librarian to manually reorder/remove queue entries and maintain an audit trail.

2. User Stories, Priorities & Story Points

Epic	User Story	Priority	Points
Book Reservation & Holding Queue	Search ISBN Repository	High	5
Book Reservation & Holding Queue	Place Book Hold	High	5
Book Reservation & Holding Queue	Join FIFO Waiting Queue	High	5
Reservation Notifications	Book Availability Notification	High	5
Reservation Notifications	Queue Position Notification	Medium	3
Fine Management	Calculate Overdue Fine	Medium	5
Fine Management	Apply Fine Limit	Medium	3
Hold Cancellation	Cancel Pending Hold	Medium	3
Hold Cancellation	Update Queue After Cancellation	Medium	5
Librarian Queue Administration	View Holding Queue	Low	3
Librarian Queue Administration	Modify Queue Entry	Low	8
Librarian Queue Administration	Record Queue Modification	Low	5
Librarian Queue Administration	Notify Affected Members	Low	5

Total backlog: 60 Story Points.

3. Sprint Plan

Sprint 1

Duration: 1 week

Sprint Goal: Implement the core book reservation and FIFO holding-queue functionality.

User Story	Points
Search ISBN Repository	5
Place Book Hold	5
Join FIFO Waiting Queue	5
Book Availability Notification	5
Cancel Pending Hold	3
Update Queue After Cancellation	5
Total	28

Sprint 2

Duration: 1 week

Sprint Goal: Implement fine management and librarian administrative queue management.

User Story	Points
Calculate Overdue Fine	5
Apply Fine Limit	3
Queue Position Notification	3
View Holding Queue	3
Modify Queue Entry	8
Record Queue Modification	5
Notify Affected Members	5
Total	32

	DB-6	Search ISBN Repository	Book Reservation ...	To Do ▾	5		
☐		DB-7	Place Book Hold	Book Reservation ...	To Do ▾	5	...
	DB-8	Join FIFO Waiting Queue	Book Reservation ...	To Do ▾	5		
	DB-9	Book Availability Notification	Reservation Notifi...	To Do ▾	5		
	DB-10	Queue Position Notification	Reservation Notifi...	To Do ▾	3		
	DB-11	Calculate Overdue Fine	Fine Management	To Do ▾	5		
	DB-12	Apply Fine Limit	Fine Management	To Do ▾	3		
	DB-13	Cancel Pending Hold	Hold Cancellation	To Do ▾	3		
	DB-14	Update Queue After Cancellation	Hold Cancellation	To Do ▾	5		
	DB-15	View Holding Queue	Librarian Queue A...	To Do ▾	3		
	DB-16	Modify Queue Entry	Librarian Queue A...	To Do ▾	8		
	DB-17	Record Queue Modification	Librarian Queue A...	To Do ▾	5		
	DB-18	Notify Affected Members	Librarian Queue A...	To Do ▾	5		

Search board



Filter

Group

Complete sprint



To Do 1

Update Queue After
Cancellation

☐ Hold Cancellation ▾

DB-14

5



+

In Progress 5

Search ISBN Repository

Book Reservation & Holding Queue ▾

DB-6

5



Place Book Hold

Book Reservation & Holding Queue ▾

DB-7

5



Join FIFO Waiting Queue

Book Reservation & Holding Queue ▾

DB-8

5



Book Availability Notification

Reservation Notifications ▾

+

Done 0



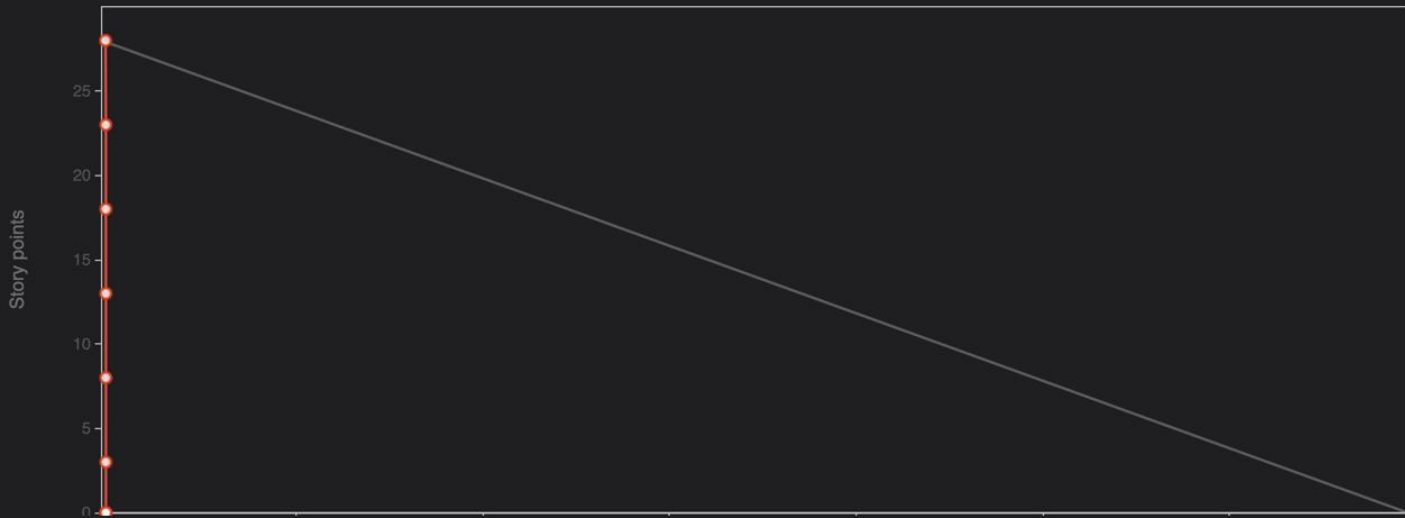
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Remaining work
Number of story points left to complete this sprint

Guideline
Ideal burn rate



To Do 4

View Holding Queue

Librarian Queue Administration ▾

DB-15 3 

Modify Queue Entry

Librarian Queue Administration ▾

DB-16 8 

Record Queue Modification

Librarian Queue Administration ▾

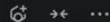
DB-17 5 

Notify Affected Members

Librarian Queue Administration ▾

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In Progress 3



Queue Position Notification

Reservation Notifications ▾

DB-10 3 

Calculate Overdue Fine

Fine Management ▾

DB-11 5 

Apply Fine Limit

Fine Management ▾

DB-12 3 

+

Done 0

+

+

Sprint

Sprint 2



Estimation field

Story points



Remaining work

Number of story points left to complete this sprint

Guideline

Ideal burn rate

Story points

30
25
20
15
10
5
0

Date



5. Reflection

Q1. Did your estimations reflect the actual effort?

Yes. The Story Point estimates reasonably reflected the actual effort. Simple operations such as viewing the holding queue were assigned 3 points, while more complex operations such as modifying and re-sequencing the queue were assigned 8 points because they involve multiple steps and greater uncertainty.

Q2. Was your backlog well-prioritized?

Yes. The backlog was prioritized according to the requirements from Lab 1. Core reservation and notification functionality was given High priority, fine management and cancellation were given Medium priority, and administrative queue management was given Low priority.

Q3. How did your simulated sprint align with your plan?

The Sprint 1 goal was to implement the core reservation and FIFO queue functionality. The selected stories were progressed from To Do to In Progress and then Done. The sprint therefore aligned with the planned objective and focused on the highest-priority functionality.

Q4. What insights did the burndown chart give about your team's capacity?

The Burndown Chart showed how the remaining Story Points decreased as User Stories were completed. It helped determine whether the team was progressing at the expected rate and provided an indication of the amount of work the team could realistically complete in a sprint.